

Your Full Name

✉ you@university.edu | ☎ +1 (555) 000-0000 | yoursite.com | github.com/yourhandle | City, State

Research interests: Numerical Analysis · Scientific Computing · Stochastic Processes · Optimization · Mathematical Modeling

EDUCATION

Ph.D. in Applied Mathematics *University of Somewhere*

Expected May 2026

Advisor: Prof. Jane Doe | GPA: 3.95/4.0

- Dissertation: "Adaptive Multilevel Methods for Stochastic PDEs on Complex Domains"
- Coursework: Numerical PDEs Functional Analysis Stochastic Calculus Optimization ML Theory

B.S. Mathematics (Honors), Minor Computer Science *State University*

May 2020

Summa Cum Laude | GPA: 3.98/4.0

- Senior thesis: "Finite Element Approximations to the Navier–Stokes Equations"

RESEARCH EXPERIENCE

Graduate Research Assistant *Computational Science Lab, University of Somewhere*

Aug 2020 – Present

Advisor: Prof. Jane Doe

- Developed adaptive multilevel Monte Carlo (MLMC) for elliptic SPDEs achieving $\mathcal{O}(h^2\varepsilon^{-2})$ complexity — $3\times$ speedup over standard MLMC on benchmark problems.
- Implemented parallel C++/MPI solver on unstructured meshes; near-linear weak scaling to 512 cores on university HPC cluster.
- Applied sparse-grid collocation for UQ in subsurface flow models across 20-dimensional parameter spaces.

Research Intern — Mathematics of Machine Learning *National Lab or Industry Partner*

Jun – Aug 2022

Mentor: Dr. John Smith

- Proved sharp $\mathcal{O}(k^{-1/2})$ SGD convergence under (L, σ) -smooth assumptions; derived matching lower bounds.
- Benchmarked quasi-Newton methods for large-scale logistic regression, outperforming L-BFGS by 18% on standard NLP datasets.

REU Participant — Dynamical Systems *Another University*

May – Jul 2019

Mentor: Prof. Alice Brown

- Computer-assisted proof of Hopf bifurcation in a 3-species Lotka–Volterra model using interval arithmetic.

PUBLICATIONS & PREPRINTS

Your Name, Jane Doe. "Adaptive Multilevel Monte Carlo for Stochastic Elliptic Equations." **SIAM Journal on Scientific Computing**, 2024. [\[link\]](#)

Your Name, John Smith. "Sharp Convergence Rates for Nonconvex SGD with Variance Reduction." **arXiv preprint arXiv:2401.00000**, 2023. [\[link\]](#)

Your Name, Alice Brown. "Computer-Assisted Proof of Hopf Bifurcation in a Predator–Prey System." **Journal of Mathematical Biology**, 2020.

TALKS & PRESENTATIONS

SIAM Annual Meeting	"Adaptive MLMC for SPDEs" (contributed talk)	2024
Applied Math Seminar	University of Somewhere (invited)	2023
ICIAM	"Nonconvex SGD convergence rates" (poster)	2023
AMS Spring Sectional	"Hopf bifurcations via interval arithmetic" (poster)	2020

TEACHING EXPERIENCE

Teaching Assistant — Numerical Analysis (MATH 5XX) *University of Somewhere*

Fall 2021, Spring 2023

Lead sections, write & grade problem sets, hold office hours for 45 students

Instructor of Record — Calculus II (MATH 1XX) *University of Somewhere*

Summer 2022

Sole instructor; course evaluations 4.7/5.0

TECHNICAL SKILLS

Languages: Python (NumPy, SciPy, JAX), C/C++, MATLAB, Julia, R, Bash

HPC & Parallel: MPI, OpenMP, SLURM, GPU computing (CUDA basics)

Math Software: FEniCS, PETSc, deal.II, Firedrake, LAPACK/BLAS

ML / Data: PyTorch, scikit-learn, pandas, Jupyter

Tools: Git, L^AT_EX, Linux, Docker, CMake

Methods: FEM, FDM, Spectral Methods, MLMC, Optimization, Bayesian Inference

HONORS, FELLOWSHIPS & AWARDS

NSF Graduate Research Fellowship

2021–2026

SIAM Student Paper Prize (honorable mention)

2023

Dean's Doctoral Excellence Award, University of Somewhere

2022

Phi Beta Kappa, State University

2020

Barry Goldwater Scholarship

2019

SERVICE & PROFESSIONAL ACTIVITIES

Referee: *SIAM J. Numer. Anal.*, *Math. Comp.*, *J. Comput. Phys.* · Organizer: Graduate Applied Math Seminar, 2022–2024 ·

Member: SIAM, AMS